a

Energy

Number frequency

0.30
0.25
0.20
0.15
0.10
0.05
0.00

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Shallow, merged wells:
approximate readout

Deep, isolated wells:
exact readout

Evidence axis x

b

Readout entropy

$y_0=2$
$H=0.02$

1 2 3

$y_0=6$
$H=1.15$

5 6 7 8 9

$y_0=12$
$H=2.45$

10 12 14

Posterior distribution
over input y_0

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

c

Predicted RT (s)

3.0
2.5
2.0
1.5

Basal latency t_0
Attractor time $\alpha E[T]$
Readout uncertainty $\beta_H E[H]$

Slow, approximate

Fast, exact

$\mathrm{RT} = t_0 + \alpha E[T] + \beta_H E[H]$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Presented numerosity n